

Annotation Transfer Tool

<https://github.com/aiyshwary/Annotation-Transfer-Tool>

OVERVIEW

A smart auto-annotation pipeline that leverages deep image similarity to transfer labels from previously human-labeled images to unannotated ones, significantly reducing manual labeling effort in computer vision workflows.

IMPACT

Reduced manual annotation effort by automatically transferring labels from existing human-labeled images using deep image similarity and a modular 8-step pipeline integrated with CVAT and AWS S3.

KEY DETAILS

1. Downloads labeled images from AWS S3 and extracts annotations from Datamano JSON format.
2. Generates image crops, normalizes labels, and creates masks for each labeled region.
3. Uses DeepImageSearch (GPU-accelerated) to identify the most visually similar labeled images for each unlabeled image.
4. Transfers and corrects annotations from matched images, then exports back to CVAT-compatible JSON format.
5. Includes normalized and denormalized visualization scripts and an interactive Jupyter demo notebook.
6. Uses VGG19 as the feature extraction backbone with FAISS indexing for similarity search; supports incremental learning where new indexed images are not re-processed.
7. Annotation transfer selects the best match per crop by minimum cosine distance, replacing class and bounding box coordinates.

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8. Automates CVAT task creation via REST API — programmatically creates cloud storage connection manifests with per-image checksums, and uploads Datumaro ZIP annotations.